Inverted MAML Card 1: Round 10 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.055, L2=1920, Mom=0.85, Clamp=±75 LSB

1. Round 10 Summary & Cumulative Comparison (R1–R9)

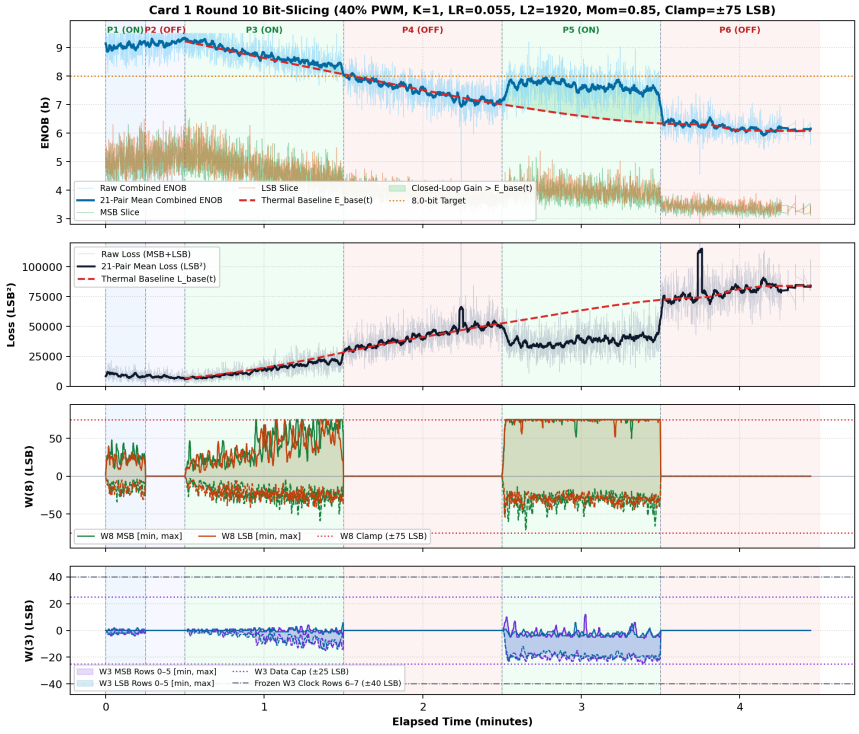
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 10 Detrended Trend Metrics:** Lift vs E_base = **+0.81 b**, Loss Ratio = **1.59x**, Climb = **+0.66 b**, Jitter = **0.38 b**, Recovery = **0.50**, Score = **+1.756**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R2dc_fix	0.035	1536	0.85	±75	+0.99b	1.69x	+1.16b	0.39b	0.00	+1.957
R3dc_fix	0.02	2560	0.8	±70	+0.04b	1.00x	+0.24b	0.34b	1.00	+0.204
R4dc_fix	0.05	1792	0.85	±70	+1.08b	1.78x	+0.97b	0.40b	0.50	+2.480
R5dc_fix	0.055	1792	0.85	±85	+1.12b	1.77x	+1.37b	0.42b	0.00	+2.393
R6dc_fix	0.055	1792	0.8	±90	+0.99b	1.72x	+1.02b	0.40b	0.00	+1.770
R7dc_fix*	0.055	2048	0.85	±75	+1.04b	1.74x	+1.17b	0.40b	0.50	+2.665
R8dc_fix	0.055	2048	0.85	±75	+1.16b	1.85x	+0.68b	0.42b	0.00	+1.855
R9dc_fix	0.055	1920	0.85	±75	+0.81b	1.59x	+0.66b	0.38b	0.50	+1.756

2. Phase-by-Phase Telemetry (4,041 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	311	8.96->9.09 (9.05b)	-	9606->8669	[48,40]	4	311
P2	Cold / Regler OFF	297	9.12->9.22 (9.17b)	-	9135->7015	[0,0]	0	0
P3	Warm / Regler ON	1068	9.28->8.46 (8.76b)	+0.19 b	6208->19392	[75,75]	15	1068
P4	Warm / Regler OFF	946	8.02->7.15 (7.49b)	-0.02 b	30433->50326	[0,0]	0	0
P5	Warm / Regler ON	819	7.20->7.56 (7.65b)	+1.14 b	47816->40070	[75,75]	25	819
P6	Warm / Regler OFF	600	6.38->6.11 (6.23b)	+0.02 b	73475->82846	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 10

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±75 LSB, while W(3) rows 0–5 stay within ±25 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 10 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

